ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals paper 1 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording.

SECTION A

QUESTION 1

1 1.1 A ✓ 1.2 C ✓ 1.3 B ✓ 1.4 D ✓ 1.5 A ✓ 1.6 B ✓ 1.7 C ✓ 1.8 D ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 True ✓ 2.2 False ✓ 2.3 True ✓ 2.4 False ✓ 2.5 True ✓ 2.6 False ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

[10]

QUESTION 3

3 3.1 C ✓ 3.2 G ✓ 3.3 I ✓ 3.4 E ✓ 3.5 A ✓ 3.6 D ✓ 3.7 F ✓ 3.8 B ✓ 3.9 J ✓ 3.10 H ✓ (10)

[10]

QUESTION 4

4 4.1 passive ✓ 4.2 sound ✓ 4.3 electrical ✓ 4.4 anode ✓ 4.5 servo ✓ (5)

[5]

TOTAL SECTION A: [35]

SECTION B

QUESTION 5

5.1 5.1.1 Breadboard ✓ 5.1.2 Resistor ✓ 5.1.3 Breakout board ✓ 5.1.4 Crocodile-clip lead ✓ 5.1.5 LED (light-emitting diode) ✓ (5)

5.2 Components can be connected quickly without soldering ✓; the circuit can easily be changed and tested ✓; components can be removed and reused ✓. (3)

5.3 A sensor detects a physical quantity and changes it into an electrical signal for the controller ✓ – e.g. ultrasonic sensor, LDR, PIR ✓. An actuator receives a signal from the controller and produces movement or another action ✓ – e.g. motor, servo, buzzer, solenoid ✓. (4)

[12]

QUESTION 6

6.1 Python ✓; the RPi.GPIO library ✓ (2)

6.2 It switches on the internal pull-up resistor ✓, so the input is HIGH when the button is not pressed ✓; the button connects the pin to GND, so it reads LOW when pressed ✓. (3)

6.3 It adds 1 to the variable presses ✓✓, counting one more visitor ✓. (3)

6.4 3 beeps ✓; each beep takes 0,2 + 0,2 = 0,4 s, so 3 × 0,4 = 1,2 s ✓, plus 1 s = 2,2 s per press ✓. (3)

6.5 It gives the visitor time to release the button ✓, so one press is not counted many times ✓ (debouncing / avoids repeated beeping) ✓. (3)

6.6 A KeyboardInterrupt happens ✓; the except block prints "Doorbell stopped" with the total number of visitors ✓; the finally block then cleans up the GPIO pins and the program ends ✓. (3)

6.7 It resets all the GPIO pins that were used back to inputs / their default state ✓✓, so that no pin is left on and the pins are safe to use in the next program ✓. (3)

6.8 Any TWO components with uses ✓✓✓ (3 marks for two well-explained), e.g. an LED that flashes for people who cannot hear the buzzer; a camera module to take a photo of the visitor; a PIR sensor to detect someone at the door; an LCD to show the number of visitors; a Wi-Fi message to a phone. (3)

[23]

QUESTION 7

7.1 The opposition that a material offers to the flow of electric current ✓, measured in ohms (Ω) ✓. (2)

7.2 $\mathrm{I} = \mathrm{V} \div \mathrm{R}$ ✓ $= 9 \div 450$ ✓ $= 0{,}02\ \mathrm{A}\ (20\ \mathrm{mA})$ ✓ (3)

7.3 $\mathrm{V} = \mathrm{I} \times \mathrm{R}$ ✓ $= 0{,}5 \times 24$ ✓ $= 12\ \mathrm{V}$ ✓ (3)

7.4 $\mathrm{P} = \mathrm{V} \times \mathrm{I}$ ✓ $= 5 \times 2$ ✓ $= 10\ \mathrm{W}$ ✓ (3)

7.5 $\mathrm{R} = \mathrm{V}^2 \div \mathrm{P}$ ✓ $= 12^2 \div 36$ ✓ $= 4\ \Omega$ ✓ $(\text{or } \mathrm{I} = 36 \div 12 = 3\ \mathrm{A}, \mathrm{R} = 12 \div 3 = 4\ \Omega)$ (3)

7.6 $\mathrm{R_T} = 100 + 220$ ✓ $= 320\ \Omega$ ✓; $\mathrm{I} = 9 \div 320$ ✓ $= 0{,}03\ \mathrm{A}\ (28{,}13\ \mathrm{mA})$ ✓ (4)

7.7 In a series circuit there is only one path, so the same current flows through all components ✓; in a parallel circuit there are several branches, each with the same voltage, and the currents add up ✓. (2)

[20]

QUESTION 8

8.1 A robot may not injure a human being ✓ or, through inaction ✓, allow a human being to come to harm ✓. (3)

8.2 No – the robot should not lock the gate yet ✓. The Second Law says the robot must obey orders from humans ✓, but only when those orders do not conflict with the First Law ✓. Locking the child out in the cold could cause the child harm ✓, and the First Law says the robot may not, through inaction, allow a human to come to harm ✓. So the robot should first help the child inside (or tell the man that the child is outside) ✓ and lock the gate once everyone is safe ✓. (7)

[10]

TOTAL SECTION B: [65]

TOTAL: 100

END OF MARKING GUIDELINE